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1 • CREATE DATABASE BankDB;
2
3 • USE BankDB;
4
5 • CREATE TABLE Customer (
6     Customer_ID INT PRIMARY KEY,
7     Customer_Name VARCHAR(100) NOT NULL,
8     Phone VARCHAR(15),
9     Email VARCHAR(100),
10    City VARCHAR(50)
11 );
12
13 • CREATE TABLE Account (
14     Account_No INT PRIMARY KEY,
15     Customer_ID INT,
16     Account_Type VARCHAR(20),
17     Balance DECIMAL(12,2) DEFAULT 0,
18     Branch VARCHAR(50),
19     FOREIGN KEY (Customer_ID)
20     REFERENCES Customer(Customer_ID)
21 );

46 (101, 'Hasini', '9876543210', 'hasini@gmail.com', 'Hyderabad'),
47 (102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
48 (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
49 (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
50 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
51
52 • INSERT INTO Account
53 (Account_No, Customer_ID, Account_Type, Balance, Branch)
54 VALUES
55 (10001, 101, 'Savings', 50000, 'Hyderabad'),
56 (10002, 102, 'Savings', 75000, 'Vijayawada'),
57 (10003, 103, 'Current', 120000, 'Bangalore'),
58 (10004, 104, 'Savings', 45000, 'Chennai'),
59 (10005, 105, 'Current', 90000, 'Hyderabad');
60
61 • INSERT INTO Bank_Transaction
62 (Account_No, Transaction_Type, Amount)
63 VALUES
64 (10001, 'DEPOSIT', 10000),
65 (10002, 'DEPOSIT', 15000),

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74      (502, 102, 'Education Loan', 1000000, 6.5),
75      (503, 103, 'Car Loan', 800000, 8.2),
76      (504, 104, 'Personal Loan', 500000, 10.5);
77
78 •   SELECT * FROM Customer;
79
80 •   SELECT * FROM Account;
81
82 •   SELECT * FROM Bank_Transaction;
83
84 •   SELECT * FROM Loan;
85
86   DELIMITER //
87
88 •   CREATE PROCEDURE GetAllCustomers()
89   BEGIN
90       SELECT * FROM Customer;
91   END //
92
93   DELIMITER ;
94
95 •   CALL GetAllCustomers();

276 •   CREATE TRIGGER PreventInsufficientWithdrawal
277   BEFORE INSERT ON Bank_Transaction
278   FOR EACH ROW
279   BEGIN
280       DECLARE CurrentBalance DECIMAL(12,2);
281
282       SELECT Balance
283       INTO CurrentBalance
284       FROM Account
285       WHERE Account_No = NEW.Account_No;
286
287       IF NEW.Transaction_Type = 'WITHDRAW'
288       AND NEW.Amount > CurrentBalance THEN
289           SIGNAL SQLSTATE '45000'
290           SET MESSAGE_TEXT =
291               'Withdrawal failed: Insufficient balance';
292       END IF;
293   END //

```

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SELECT Balance
INTO SenderBalance
FROM Account
WHERE Account_No = SenderAccount;

IF SenderBalance < TransferAmount THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT =
        'Transfer failed: Insufficient balance';
ELSE
    UPDATE Account
    SET Balance = Balance - TransferAmount
    WHERE Account_No = SenderAccount;

    UPDATE Account
    SET Balance = Balance + TransferAmount
    WHERE Account_No = ReceiverAccount;
END IF;
END //

385     SELECT Balance
386     INTO p_Balance
387     FROM Account
388     WHERE Account_No = p_Account_No
389     END //
390
391     DELIMITER ;
392
393 • CALL GetBalance(10001, @CurrentBala
394
395 • SELECT @CurrentBalance;
396
397 • SHOW TRIGGERS;
398
399 • SHOW CREATE TRIGGER CheckBalance;
400
401 • DROP TRIGGER CheckBalance;
402
403 • SHOW PROCEDURE STATUS
404     WHERE Db = 'BankDB';
405
406 • DROP PROCEDURE GetAllCustomers;

```

75	14:38:54	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCR...	0 row(s) affected	0.031 sec
76	14:38:54	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR...	0 row(s) affected	0.016 sec
77	14:38:54	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALU...	1 row(s) affected	0.000 sec
78	14:38:54	SELECT * FROM Transaction_Audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
79	14:38:54	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Tr...	0 row(s) affected	0.016 sec
80	14:38:54	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALU...	1 row(s) affected	0.000 sec
81	14:38:54	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
82	14:38:54	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON Bank_Tr...	0 row(s) affected	0.015 sec
83	14:38:54	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALU... Error Code: 1644. Withdrawal failed: Insufficient balance	0 row(s) affected	0.000 sec
84	14:38:54	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN Receiver...	0 row(s) affected	0.016 sec
85	14:38:54	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected	0.000 sec
86	14:38:54	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
87	14:38:54	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN ...	0 row(s) affected	0.015 sec
88	14:38:54	CALL GetCustomerLoans(101)	1 row(s) returned	0.000 sec / 0.000 sec
89	14:38:54	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(1...	0 row(s) affected	0.000 sec
90	14:38:54	CALL HighBalanceAccounts(50000)	4 row(s) returned	0.000 sec / 0.000 sec
91	14:38:54	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance...	0 row(s) affected	0.000 sec
92	14:38:54	CALL GetBalance(10001, @CurrentBalance)	1 row(s) affected	0.016 sec
93	14:38:54	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
94	14:38:54	SHOW TRIGGERS	5 row(s) returned	0.016 sec / 0.000 sec
95	14:38:54	SHOW CREATE TRIGGER CheckBalance	1 row(s) returned	0.000 sec / 0.000 sec
96	14:38:54	DROP TRIGGER CheckBalance	0 row(s) affected	0.016 sec
97	14:38:54	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	9 row(s) returned	0.000 sec / 0.000 sec
98	14:38:54	DROP PROCEDURE GetAllCustomers	0 row(s) affected	0.015 sec

101	Hasini	9876543210	hasini@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
NULL	NULL	NULL	NULL	NULL

Customer_ID	Customer_Name	Phone	Email	City
101	Hasini	9876543210	hasini@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
▶	101	Hasini	10001	Savings	62000.00	Hyderabad

	Db	Name	Type	Definer	Modified	Created	Security_type	Comment
►	bankdb	DepositMoney	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	
	bankdb	GetAccountDetails	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	
	bankdb	GetAllCustomers	PROCEDURE	root@localhost	2026-08-30 15:12:24	2026-08-30 15:12:24	DEFINER	
	bankdb	GetBalance	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	
	bankdb	GetCustomerAccounts	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	
	bankdb	GetCustomerLoans	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	
	bankdb	HighBalanceAccounts	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	
	bankdb	TransferMoney	PROCEDURE	root@localhost	2026-08-30 14:38:54	2026-08-30 14:38:54	DEFINER	